

Predicting solar power generation

A solar energy data science project using various weather features to predict the expected energy output for a solar farm and to estimate net power demand by combining generation and consumption data.

Project Motivation

Accurately forecasting solar power generation and energy consumption is essential for maintaining grid stability and optimizing renewable energy integration. Reliable predictions enable utilities to balance supply and demand, reduce reliance on nonrenewable sources, and improve energy planning. This project supports these goals by modeling solar generation and estimating net power demand to enhance decision-making for sustainable energy management.

Dataset

The solar energy dataset used comes from the *PVDAQ* Solar Data Bounty Prize, focusing on site 9068 near Denver, Colorado. Some weather data were included in this dataset, while additional meteorological data were acquired from *ERA5* and *NSRDB* datasets. These datasets provide hourly measurements of solar and atmospheric conditions for Kersey, Colorado.

Modeling Approach

We began with baseline models that predicted solar power output using the mean of previous years. More advanced models, including Linear Regression, ExtraTrees, and XGBoost, were trained on 2022, validated on 2023, and tested on 2024 datasets. We experimented with different feature sets: starting with solar zenith angle, and then adding cloud type, total cloud coverage, humidity, precipitation, temperature, and lag terms. To better represent physical effects, we introduced additional features such as temperature change (ΔT) and the inverter power factor to account for thermal lag and power loss. Using data from 2021–2024, we performed K-fold cross-validation and fine-tuned ensemble model parameters with grid search and cross-validation for both XGBRegressor and XGBRFRegressor.

Results

The baseline models performed poorly ($R^2 \approx -0.001$), while the tuned ensemble models substantially improved predictive accuracy. The ExtraTrees and XGBoost regressors both achieved, $RMSE \approx 767 - 768$ and $R^2 \approx 0.63$. With enhanced features, the XGBRegressor triumphed with a mean RMSE of 544 $R^2 \approx 0.83$, and the XGBRFRegressor reached an RMSE of 659 ($R^2 \approx 0.75$). Among the weather features ΔT , cloud type, temperature, and zenith angle provided the greatest predictive power. Also, Nighttime data were excluded to avoid inflated performance due to zero solar output. By integrating energy consumption data, we estimated the net power demand, demonstrating the model's capability to forecast both production and real-time grid balance.

Future direction

Future work will focus on incorporating energy consumption data to estimate net power demand, expanding to additional solar farm sites, and adding new predictive features such as panel tilt or long-term weather trends. With more extensive data and fine-tuned hyperparameters, model accuracy and generalizability can be further improved.